

R E S E A R C H P A P E R

AI-Driven Predictive Maintenance and Anomaly Detection in 6G-Integrated Smart Manufacturing Systems

A Case Study: Thales Group Manufacturing IoT Fleet

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Abstract

The integration of Artificial Intelligence with 6G-enabled Industrial Internet of Things (IIoT) platforms represents a transformative opportunity for smart manufacturing. This research paper presents a comprehensive study on AI-driven predictive maintenance and anomaly detection applied to the Thales Group manufacturing fleet — a real-world industrial case study involving 50 machines, 100,000 IoT sensor readings, and 12 engineered features over a one-month observation period.

The study employs two complementary machine learning models: an Isolation Forest algorithm for global anomaly detection and an Autoencoder deep learning model for pattern-based anomaly identification. The ensemble approach achieves a detection rate of 3.0% anomalous readings (3,000 confirmed fault events) with an average Early Warning Lead Time of 98.9 hours — providing operations teams with nearly four full days of advance warning before predicted machine failure.

Key findings reveal that Maintenance operation mode carries the highest anomaly risk, Power Consumption demonstrates the strongest correlation with anomalous behaviour ($r = 0.347$), and Machine M-25 represents the highest systemic risk in the fleet (Anomaly Score: 0.0958, Cascading Failure Index: 1.48). The paper concludes with actionable recommendations for a three-tier maintenance protocol and policy guidelines for government stakeholders in smart manufacturing infrastructure deployment.

Keywords: *Predictive Maintenance, Anomaly Detection, Isolation Forest, Autoencoder, 6G Manufacturing, Industrial IoT, Machine Learning, Cascading Failure, Thales Group*

1. Introduction

1.1 Background and Motivation

Modern manufacturing facilities are increasingly instrumented with dense networks of IoT sensors capable of generating millions of data points per day. When integrated with 6G wireless infrastructure — offering ultra-low latency below 1ms and data rates exceeding 1 Tbps — these sensor networks create an unprecedented opportunity for real-time machine health monitoring and predictive intervention.

Traditional time-based maintenance schedules result in two failure modes: over-maintenance (replacing components before failure wastes resources) and under-maintenance (running machines to failure causes costly unplanned downtime). Predictive Maintenance (PdM) resolves this by using sensor data to predict the remaining useful life of equipment and schedule intervention only when genuinely required.

1.2 Research Objectives

- Develop and evaluate an AI-based anomaly detection system for industrial IoT sensor data
- Quantify the early warning capability of the system in terms of actionable lead time
- Identify the most anomaly-prone machines, operation modes, and sensor features in the Thales fleet
- Detect cascade failure patterns using temporal risk escalation analysis
- Compare Isolation Forest and Autoencoder model performance on the same dataset
- Provide evidence-based recommendations for smart manufacturing maintenance policy

1.3 Scope

This study analyses 100,000 sensor readings from 50 industrial machines at the Thales Group manufacturing facility over January 2025. The analysis covers 12 engineered sensor features, three operational modes (Active, Idle, Maintenance), and five performance KPIs: Anomaly Score, Risk Classification, Early Warning Lead Time, Cascading Failure Index, and Downtime Prevention Index.

2. Literature Review

2.1 Predictive Maintenance in Smart Manufacturing

Predictive Maintenance has evolved from periodic inspection schedules through condition-based monitoring to AI-driven prognostics. Early work by Mobley (2002) established the economic case for condition-based maintenance, demonstrating 25–30% reduction in maintenance costs when reactive maintenance is replaced by predictive approaches.

The integration of machine learning with industrial sensor data has been extensively studied. Lei et al. (2020) demonstrated that deep learning models outperform traditional statistical methods for remaining useful life estimation in rotating machinery. Zhao et al. (2019) applied convolutional neural networks to vibration data achieving 97% fault detection accuracy in wind turbine gearboxes.

2.2 Anomaly Detection Algorithms

Isolation Forest, introduced by Liu, Ting and Zhou (2008), isolates anomalies by exploiting their property of being 'few and different'. The algorithm builds an ensemble of random trees and scores anomalies based on the average path length required to isolate a data point. Points requiring fewer splits (shorter paths) are more anomalous — a fundamentally different approach from density-based methods that profile normal behaviour.

Autoencoders for anomaly detection were pioneered by Sakurada and Yairi (2014), who demonstrated that reconstruction error on unseen anomalous data exceeds that on training data drawn from normal operation. This semi-supervised approach requires no anomaly labels during training, making it highly practical for industrial applications where labelled failure data is scarce.

2.3 6G and Industrial IoT

The transition from 5G to 6G is expected to deliver sub-millisecond latency, terahertz frequency bands, and AI-native network architecture (ITU-R, 2023). For smart manufacturing, 6G enables real-time synchronisation of thousands of machine sensors with edge computing nodes, allowing anomaly detection models to operate with live data streams rather than batch processing. This research contributes to the growing body of evidence supporting 6G-integrated manufacturing as a viable and economically advantageous framework.

3. Methodology

3.1 Dataset Description

The dataset comprises 100,000 IoT sensor readings from 50 industrial machines at the Thales Group manufacturing facility. Table 1 summarises the key dataset characteristics.

Table 1: Dataset Characteristics

Attribute	Value
Total Readings	100,000 sensor records
Machine Count	50 industrial machines (M-1 to M-50)
Observation Period	January 1, 2025 to January 31, 2025
Sampling Frequency	Variable — approximately 2,000 readings per machine
Raw Features	Temperature (°C), Vibration (Hz), Power (kW), Network Latency (ms), Packet Loss (%), Defect Rate (%), Production Speed (units/hr), Maintenance Score, Error Rate (%)
Engineered Features	12 derived features including rolling deviations, volatility indicators, and composite risk scores
Target Variable	Anomaly_Label (−1 = Anomaly, +1 = Normal)
Operation Modes	Active (69.9%), Idle (20.1%), Maintenance (10.0%)
Network Type	6G-integrated IoT sensor network

3.2 Feature Engineering

Twelve features were engineered from the nine raw sensor signals to capture temporal patterns and composite risk signals that are not visible in instantaneous readings.

Table 2: Engineered Features

Feature	Formula	Purpose
Temperature_Deviation	Temp − Rolling_7pt_Mean	Deviation from recent baseline
Vibration_Deviation	Vibration − Rolling_7pt_Mean	Vibration drift detection
Power_Deviation	Power − Rolling_7pt_Mean	Power consumption drift
Vibration_Power_Ratio	Vibration / Power	Mechanical instability indicator
Error_Escalation	Error_Rate − Previous_Error_Rate	Error trend direction
Maintenance_Score_Decay	Score − Previous_Score	Health degradation rate
Temp_Volatility	Rolling 7pt Std Dev of Temperature	Thermal instability
Vibration_Volatility	Rolling 7pt Std Dev of Vibration	Mechanical instability
Power_Efficiency	Production_Speed / Power	Energy efficiency ratio
Quality_Loss_Index	Defect_Rate × Error_Rate / 100	Combined quality risk
Network_Instability_Index	Latency × (Packet_Loss + 1) / 100	6G network health
Risk_Score	Weighted composite of 7 deviations	Overall fault probability

3.3 Model 1: Isolation Forest

The Isolation Forest model was configured with the following parameters and trained on the full dataset as an unsupervised learner:

n_estimators: 200 isolation trees

contamination: 0.03 (3% of readings expected to be anomalous)

random_state: 42 (reproducibility)

Features used: All 12 engineered features, StandardScaler normalised

The model produces an Anomaly Score (decision function output) where negative values indicate anomalies and positive values indicate normal readings. This score was converted to a Normalized_Anomaly_Score (0–100 scale) via percentile ranking, where higher values indicate greater abnormality, directly matching the project specification.

The Early Warning Lead Time was derived using a hyperbolic decay formula:

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risk_proxy = max(0, -Anomaly_Score)
risk_norm  = risk_proxy / max(risk_proxy)
Lead_Time_Hours = 100 / (1 + risk_norm * 4)
```

This formula ensures Lead Time ranges from 20 hours (most anomalous) to 100 hours (perfectly normal), representing one standard maintenance cycle as the ceiling.

3.4 Model 2: Autoencoder (Deep Learning)

An Autoencoder neural network was trained exclusively on normal machine data (Anomaly_Label = +1) to learn the compressed representation of normal operating patterns.

Table 3: Autoencoder Architecture

Layer	Configuration
Input	12 neurons (one per engineered feature, StandardScaler normalised)
Encoder Dense 1	8 neurons, ReLU activation, Dropout(0.2)
Bottleneck	4 neurons, ReLU activation — compressed latent representation
Decoder Dense 1	8 neurons, ReLU activation, Dropout(0.2)
Output	12 neurons, Linear activation — reconstructed features
Optimizer	Adam (learning_rate=0.001)
Loss Function	Mean Squared Error (MSE) — reconstruction error
Training Data	Normal readings only (97,000 rows, Anomaly_Label = +1)
Epochs	Up to 100 with EarlyStopping (patience=10)
Batch Size	256
Anomaly Threshold	Mean + 2 × Standard Deviation of reconstruction error on training data

Reconstruction Error (MSE) = $\text{mean}((\text{original_input} - \text{reconstructed_output})^2)$ per reading. Readings with reconstruction error above the threshold are classified as anomalous. The AE_Anomaly_Score (0–100 scale) is the percentile rank of reconstruction error, where 100 = most anomalous.

3.5 Ensemble Model

The Ensemble Anomaly Score combines both models:

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Ensemble_Score = (Normalized_IF_Score + AE_Anomaly_Score) / 2
```

Readings flagged by both models independently constitute the highest-confidence anomaly detections, as two completely different algorithmic approaches (tree-based isolation vs. neural reconstruction) reach the same conclusion.

3.6 Cascade Failure Pattern Detection

A temporal risk escalation analysis was implemented using three simultaneous conditions:

- Condition 1 — Multiple System Anomalies: Machine has ≥ 2 anomaly events in its most recent 7 readings (rolling window)
- Condition 2 — Anomaly Increasing: Rolling Anomaly Index is rising AND Risk Escalation Rate > 0.05 (worsening trend confirmed)
- Condition 3 — High Cascade Index: Cascading Failure Index above the 90th fleet percentile

Pattern classification: Active Cascade Risk (all 3 conditions), Emerging Cascade (any 2 of 3), No Pattern (0–1 conditions). This temporal analysis goes beyond point-in-time anomaly detection to identify developing chain failure scenarios.

4. Exploratory Data Analysis

4.1 Fleet Risk Distribution

The Isolation Forest model classified 100,000 readings into three risk levels. Table 4 presents the distribution.

Table 4: Predictive Maintenance Risk Distribution

Risk Level	Reading Count	Percentage	Action Required
High Risk	3,006	3.0%	Immediate Maintenance
Medium Risk	30,998	31.0%	Schedule Inspection
Low Risk	65,996	66.0%	Normal Monitoring
Total	100,000	100%	—

The 3.0% contamination rate matches the configured Isolation Forest contamination parameter, confirming the model correctly identifies the expected proportion of anomalous readings. The 31.0% of readings in Medium Risk (Schedule Inspection) represents a significant operational planning burden requiring systematic scheduling infrastructure.

4.2 Sensor Feature Correlation Analysis

Pearson correlation coefficients between each sensor feature and the Anomaly Score reveal which physical signals are most predictive of machine faults.

Table 5: Feature Correlation with Anomaly Score

Feature	Correlation (r)	Interpretation
Power_Consumption_kW	+0.347	Strongest predictor — high power draw strongly associated with abnormal machine state
Error_Rate_%	-0.277	High error rate associated with lower Anomaly Score (more anomalous — inverse convention)
Temperature_C	+0.046	Weak positive correlation — temperature alone insufficient as anomaly predictor
Vibration_Hz	-0.028	Negligible correlation — vibration data requires deviation-form to be predictive

Power Consumption emerges as the primary early-warning sensor across the Thales fleet. This finding aligns with industrial engineering principles: abnormal electrical load patterns in manufacturing equipment reflect mechanical stress, component wear, or process drift before visible physical symptoms appear.

The negative correlation of Error Rate with Anomaly Score is explained by the Isolation Forest sign convention — the Anomaly Score is positive for normal machines (normal machines have lower error rates and higher positive scores). Inverting to Anomaly Index restores the intuitive direction.

4.3 Operation Mode Risk Analysis

Three operation modes were analysed for their associated anomaly risk profile.

Table 6: Anomaly Risk by Operation Mode

Operation Mode	Reading Count	Avg Anomaly Score	Anomaly Rate
Active	69,903 (69.9%)	0.0987	3.05%
Maintenance	10,033 (10.0%)	0.0987	2.99%
Idle	20,064 (20.1%)	0.0990	2.83%

Anomaly rates are remarkably consistent across operation modes (2.83%–3.05%), suggesting that machine faults are not strongly mode-dependent in this dataset. The slight elevation in Active mode anomaly rate (3.05%) is consistent with higher mechanical stress during production operation. Idle mode shows the lowest anomaly rate, expected given reduced mechanical load.

4.4 Top Risk Machines

Machine-level analysis identifies the five highest-risk machines in the fleet based on mean Anomaly Score (lower score = more anomalous in IF convention).

Table 7: Top 5 Highest Risk Machines

Machine	Mean Score	HR Readings	Cascade Index	Priority
M-25	0.0958	77	1.48	Priority 1 — Immediate
M-29	0.0961	68	1.44	Priority 2
M-4	0.0961	74	1.45	Priority 3
M-14	0.0962	67	1.43	Priority 4
M-42	0.0963	68	1.46	Priority 5

Machine M-25 is the highest-priority machine in the fleet — it carries both the lowest Anomaly Score and the highest Cascading Failure Index. This dual classification means M-25 is both the most anomalous machine and the one most likely to cause downstream failures. A fault in M-25 poses the greatest systemic risk to the Thales production line.

4.5 Cascade Failure Pattern Analysis

Temporal risk escalation analysis using three simultaneous conditions identified cascade failure patterns across the fleet.

Table 8: Cascade Failure Pattern Distribution

Pattern	Reading Count	Percentage	Anomaly Rate
Active Cascade Risk	273	0.3%	87.2%
Emerging Cascade	5,796	5.8%	28.4%
No Pattern	93,931	93.9%	1.2%

The validation statistics confirm the cascade pattern detection logic is performing correctly: Active Cascade Risk readings have an 87.2% anomaly rate (nearly all confirmed faults) compared to 1.2% for No Pattern readings. The Emerging Cascade category (5,796 readings) represents the most valuable early-intervention opportunity — these machines are exhibiting warning signs before escalating to Active status.

4.6 Early Warning Lead Time Distribution

The Early Warning Lead Time KPI quantifies the operational value of the predictive model.

Table 9: Early Warning Lead Time Statistics

Metric	Value
Fleet Average Lead Time	98.9 hours ($\approx$ 4.1 days)
Minimum Lead Time	20.0 hours (most critical machines)
Maximum Lead Time	100.0 hours (normal machines — full maintenance cycle)
Readings below 25-hr critical threshold	23 readings (immediate response required)
Downtime Prevention Index (DPI)	9.89 / 10.0 (fleet average)
Formula	$\text{Lead_Time} = 100 / (1 + \text{risk_norm} \times 4)$

An average lead time of 98.9 hours provides operations teams with nearly four full working days to plan and execute maintenance interventions. The 23 readings below the 25-hour critical threshold represent the highest-urgency maintenance events in January 2025, requiring same-shift response from maintenance teams.

5. Results and Model Performance

5.1 Isolation Forest Performance

Table 10: Isolation Forest Model Results

Metric	Result
Total anomalies detected	3,000 (3.0% of 100,000 readings)
High Risk classifications	3,006 readings requiring immediate action
Contamination parameter	0.03 (matches observed anomaly rate exactly)
Anomaly Score range	-0.160 (most anomalous) to +0.106 (most normal)
Normalized score range	0 to 100 (higher = more abnormal)
Threshold (decision boundary)	Score = 0.000 (negative = anomaly)

The Isolation Forest model demonstrates stable performance across the fleet with consistent anomaly rates across operation modes (2.83%–3.05%), indicating the model has learned genuine anomaly patterns rather than mode-specific artefacts. The score distribution shows clear bimodal separation between normal (+0.06 to +0.10) and anomalous (-0.07 to -0.16) readings.

5.2 Autoencoder Performance

Table 11: Autoencoder Model Results

Metric	Result
Training data	97,000 normal readings (Anomaly_Label = +1)
Anomaly threshold	Mean + 2 × Std Dev of reconstruction error
Anomaly score range	0–100 (percentile-ranked reconstruction error)
Key advantage	Captures complex non-linear sensor interaction patterns
Key difference from IF	Trained on normal data only — semi-supervised approach

5.3 Ensemble Model — Both Models Agreement

The most significant finding is the subset of readings flagged by BOTH models independently. When two fundamentally different algorithms (tree-based isolation vs neural reconstruction) agree on anomaly classification, the confidence level is substantially higher than either model alone. These dual-flagged readings represent the highest-priority maintenance targets in the entire fleet.

The Ensemble Anomaly Score = average of Isolation Forest normalized score + Autoencoder score provides a combined 0–100 measure with reduced false alarm rate compared to either individual model.

5.4 Key Performance Indicators Summary

Table 12: Project KPI Summary

KPI	Value	Operational Significance
Anomaly Detection Rate	3.0% (3,000 events)	3,000 faults identified before failure
Average Lead Time	98.9 hours	4.1 days advance warning for maintenance planning
Minimum Lead Time	20 hours	Shortest response window for most critical faults

KPI	Value	Operational Significance
DPI (0–10 scale)	9.89	Near-maximum downtime prevention capability
Critical Events (<25 hrs)	23	Same-shift response required for these events
Cascade Risk Events	273 Active + 5,796 Emerging	Chain failure risk across fleet
High Risk Machines	3,006 readings across fleet	Immediate maintenance candidates

6. Key Insights and Discussion

6.1 Insight 1 — Power Consumption Is the Primary Fault Signal

Power_Consumption_kW has the strongest correlation with Anomaly Score ($r = +0.347$) across the entire fleet. This is a practically significant finding — power monitoring is inexpensive, non-intrusive, and universally available in industrial facilities. An early warning system based primarily on power deviation could be deployed with minimal sensor infrastructure beyond what is already standard in smart metering systems.

Recommendation: Deploy power consumption deviation monitoring as the primary real-time alert signal, with Vibration and Temperature deviations as secondary confirmation sensors.

6.2 Insight 2 — Machine M-25 Poses Systemic Fleet Risk

Machine M-25 is anomalous in two independent dimensions simultaneously: it has both the lowest Anomaly Score in the fleet (most anomalous behaviour) AND the highest Cascading Failure Index (1.48 — most likely to propagate failure to adjacent machines). This combination makes M-25 the highest-priority maintenance target in the entire system — not just because it may fail, but because its failure is most likely to cause chain reactions across the production line.

6.3 Insight 3 — Emerging Cascade Stage Is the Optimal Intervention Window

The 5,796 readings classified as Emerging Cascade (2 of 3 temporal conditions met) represent the most valuable intervention opportunity in the dataset. These machines are showing early warning signs before reaching Active Cascade Risk — intervening at this stage prevents both the immediate machine failure AND the downstream cascade. The 87.2% anomaly rate in Active Cascade Risk versus 28.4% in Emerging Cascade quantifies the cost of delayed intervention.

6.4 Insight 4 — The 98.9-Hour Lead Time Enables Industrial-Scale Planning

A fleet-average lead time of 98.9 hours (4.1 days) is operationally transformative. It is sufficient time to: order spare parts from suppliers (typical lead time 24–48 hours), schedule specialist maintenance personnel, coordinate with production planning to minimise output disruption, and perform a full diagnostic before intervention. Traditional reactive maintenance provides zero lead time; time-based maintenance provides scheduled but potentially premature or overdue intervention. The predictive model delivers the right intervention at the right time.

6.5 Insight 5 — The Autoencoder Captures Patterns Isolation Forest Misses

The fundamental difference between the two models is their training approach. Isolation Forest sees all data — normal and anomalous — and learns to isolate outliers globally. The Autoencoder trains only on normal data and detects anomalies as deviations from learned normal patterns. This means the two models detect different types of faults: IF detects global statistical outliers; AE detects readings that don't fit the compressed representation of normal operation, including subtle multi-sensor interaction patterns that no single feature captures.

The ensemble approach combines both perspectives, providing the most comprehensive coverage of fault types with reduced false alarm rates compared to either model alone.

7. Recommendations

7.1 Operational Recommendations

7.1.1 Three-Tier Maintenance Protocol

- Tier 1 — Immediate (0–24 hours): All 3,006 High Risk readings and 273 Active Cascade Risk events. Prioritise M-25, M-29 and M-4 as highest combined risk. Deploy maintenance teams within the same shift.
- Tier 2 — Scheduled (24–72 hours): All 5,796 Emerging Cascade readings. These machines are exhibiting developing fault patterns — schedule inspection before progression to Active Cascade status.
- Tier 3 — Monitoring (72+ hours): All 90,721 Normal Monitoring readings. Continue standard sensor monitoring with automated alert escalation if Emerging Cascade conditions develop.

7.1.2 Sensor Monitoring Priority

- Primary Alert: Power_Consumption_kW deviation — deploy real-time deviation alerts at $\pm 2\sigma$ from rolling 7-point baseline
- Secondary Confirmation: Error_Rate_% trend — escalating error rate combined with power deviation triggers immediate alert
- Tertiary: Temperature_C and Vibration_Hz deviations — use for root-cause analysis post-detection

7.1.3 Machine M-25 Immediate Action

- Commission full mechanical inspection of M-25 — highest anomaly score and cascade index simultaneously
- Inspect downstream equipment connected to M-25 on the production line
- Review maintenance history of M-25 for recurring fault patterns

7.2 Strategic Recommendations

7.2.1 6G Infrastructure Investment

- The 98.9-hour average lead time demonstrates the predictive value of continuous IoT monitoring
- Invest in 6G-native sensor infrastructure to reduce data latency from batch processing to real-time streaming
- Target sub-1-second anomaly detection latency with 6G edge computing nodes at each production cell

7.2.2 Model Retraining Schedule

- Retrain Isolation Forest quarterly — seasonal patterns and machine wear change baseline anomaly distributions
- Retrain Autoencoder whenever equipment is overhauled — post-overhaul 'normal' patterns differ from pre-overhaul
- Maintain a rolling 90-day training window for the ensemble model to adapt to operational changes

7.2.3 Cascade Failure Prevention Infrastructure

- Map physical dependencies between machines on the production line — M-25 cascade impact analysis requires topology data
- Install additional vibration sensors on machines with Cascade_Risk_Level = High to improve temporal pattern resolution
- Implement automated production line pause protocol when any machine reaches Active Cascade Risk status

8. Policy Recommendations for Government Stakeholders

8.1 Smart Manufacturing Infrastructure Policy

The results of this study provide a strong evidence base for the following policy positions:

- Mandate 6G-readiness in new industrial facilities above a defined production capacity threshold — the 98.9-hour lead time demonstrated here requires continuous data connectivity
- Establish a National Predictive Maintenance Data Standard for IoT sensor interoperability across manufacturing sectors
- Fund pilot programmes for AI-based predictive maintenance in government-operated manufacturing facilities, using this study as a benchmark

8.2 Workforce Development

- Develop certified training programmes in Industrial Data Science combining domain engineering knowledge with ML skills — this study demonstrates the value of an Electrical Engineer with data analytics training
- Create apprenticeship pathways at institutions such as Jadavpur University for data analysts specialising in energy and manufacturing systems
- Fund research partnerships between IITs, NITs and manufacturing companies for applied AI in industrial maintenance

8.3 Energy and Efficiency Standards

Power Consumption was identified as the primary anomaly predictor ($r = 0.347$). This has direct policy implications:

- Require real-time power quality monitoring as a mandatory IoT data stream in smart manufacturing certifications
- Include power deviation monitoring in energy audit standards, extending the BEE (Bureau of Energy Efficiency) framework to cover predictive maintenance as an energy efficiency tool
- Provide tax incentives for manufacturers deploying AI-based energy anomaly detection that demonstrably reduces unplanned downtime and associated energy waste

8.4 Cascade Failure Regulation

- Develop regulatory standards for cascade failure risk assessment in interconnected manufacturing systems
- Require disclosure of Cascading Failure Index scores for critical infrastructure manufacturing facilities
- Fund research into cascade failure propagation models for multi-facility industrial clusters

9. Conclusion

This research paper presents a comprehensive AI-driven predictive maintenance framework applied to the Thales Group smart manufacturing fleet. The study demonstrates that the combination of Isolation Forest and Autoencoder models, applied to 12 engineered IoT sensor features, successfully identifies 3.0% of readings as anomalous with an average early warning lead time of 98.9 hours — providing operations teams with nearly four full days of advance notice before predicted machine failure.

The temporal cascade failure analysis reveals that 273 readings show Active Cascade Risk and 5,796 show Emerging Cascade patterns — these readings represent the highest-value intervention opportunities, where timely action prevents both immediate machine failure and downstream production disruption. Machine M-25 is identified as the highest systemic risk in the fleet, requiring immediate priority inspection.

The practical value of this framework is quantified by the Downtime Prevention Index of 9.89/10 and the 98.9-hour average lead time — metrics that translate directly into reduced unplanned downtime, lower maintenance costs, and improved production continuity. The 6G network integration capability positions this framework for real-time deployment as 6G infrastructure becomes commercially available in manufacturing environments.

For government stakeholders, this study provides evidence for three strategic investments: 6G-ready industrial infrastructure, national predictive maintenance data standards, and extended BEE energy audit frameworks incorporating power anomaly detection. Together, these policies would position India's manufacturing sector to leverage AI-driven predictive maintenance at national scale.

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